

Ticket A2T4

1. Purpose

This document validates that the backup and recovery system meets the RPO, RTO, and MTTR targets defined in Lab 11.1. Each in-scope service was drilled three times across a 24-hour window. Results are evidence-based and compared against defined targets.

2. Backup & Recovery Architecture Summary

Host Configuration (/etc/ trees)

Rsync-based backup of `/etc/` and `/etc/puppetlabs/` from mgmt-d, db-d, and app-d. Cron-scheduled at staggered times (00/06/12/18:00 UTC). Snapshots land under `/home/backup/<host>/` on backup-d using `--link-dest` for storage efficiency. Restore is performed via rsync into `/var/tmp/restore-test/` on the source host — never over the live tree.

MariaDB (db-d logical dumps)

`mysqldump --all-databases --single-transaction` runs on db-d at 00:45/06:45/12:45/18:45 UTC. Dumps are stored locally at `/home/backup/db-d/mariadb/` with a `latest.sql` symlink. Restore is performed on backup-d into a throwaway `drill_test` database, validated with `SELECT 1`, then dropped. db-d is never touched during drills.

OwnCloud user data (app-d)

Rsync backup of `/var/www/owncloud/config/` from app-d to `/home/backup/app-d/owncloud/` on backup-d. Restore is performed into `/var/tmp/restore-test/` on app-d. **Known limitation:** the data directory (`/var/www/owncloud/data/`) is not in backup scope, so `occ status` post-flight fails. This is a documented finding — see Section 9.

3. Test Methodology

Three drills per service, spaced across 24 hours. Both team members operated at least one drill each. Drills followed the six-stage protocol: scope → pre-conditions → inject failure → detect → recover → measure. A drill is successful if the post-flight check passes and restored data is consistent with the backup timestamp.

Restore targets used throughout:

- Host config → `/var/tmp/restore-test/` on source host
- MariaDB → `drill_test` database on backup-d
- OwnCloud → `/var/tmp/restore-test/` on app-d

4. Targets (from Lab 11.1)

Service	Target RPO	Target RTO
Host config	24 hours	10 min
MariaDB	24 hours	15 min
OwnCloud	24 hours	20 min

5. Drill Records

Service 1 — Host Configuration

Drill 1 — Operator: Ed

Field	Value
T0	13-06 03:46
Latest backup timestamp	13-06 03:41
Backup age (RPO exposure)	5 min
T_break	03:47 — <code>mv /etc/puppetlabs /etc/puppetlabs.bak</code>
T_detect	03:47 — manual observation
T_start	03:47
T_done	03:48:58
RPO measured	7 min
RTO measured	2 min
Post-flight	Files present in <code>/var/tmp/restore-test/</code>
Result	PASS

Drill 2 — Operator: Logan

Field	Value
T0	13-06 10:05
Latest backup timestamp	13-06 06:00
Backup age (RPO exposure)	4 hr 5 min
T_break	10:06 — <code>mv /etc/puppetlabs /etc/puppetlabs.bak</code>

Field	Value
T_detect	10:06 — manual observation
T_start	10:06
T_done	10:08
RPO measured	4 hr 6 min
RTO measured	2 min
Post-flight	Files present in <code>/var/tmp/restore-test/</code>
Result	PASS

Drill 3 — Operator: Ed

Field	Value
T0	13-06 20:00
Latest backup timestamp	13-06 18:00
Backup age (RPO exposure)	2 hr
T_break	20:01 — <code>mv /etc/puppetlabs /etc/puppetlabs.bak</code>
T_detect	20:01 — manual observation
T_start	20:01
T_done	20:03
RPO measured	2 hr 1 min
RTO measured	2 min
Post-flight	Files present in <code>/var/tmp/restore-test/</code>
Result	PASS

Service 2 — MariaDB

Drill 1 — Operator: Ed

Field	Value
T0	13-06 03:46
Latest backup timestamp	13-06 03:46
Backup age (RPO exposure)	~1 min
T_break	N/A — throwaway DB, no live service disrupted
T_detect	N/A

Field	Value
T_start	03:47:17
T_done	03:47:21
RPO measured	1 min
RTO measured	4 sec
Post-flight	SELECT 1 returned 1 row
Result	PASS

Drill 2 — Operator: Logan

Field	Value
T0	13-06 10:05
Latest backup timestamp	13-06 06:45
Backup age (RPO exposure)	3 hr 20 min
T_break	N/A
T_detect	N/A
T_start	10:06
T_done	10:06
RPO measured	3 hr 20 min
RTO measured	4 sec
Post-flight	SELECT 1 returned 1 row
Result	PASS

Drill 3 — Operator: Ed

Field	Value
T0	13-06 20:00
Latest backup timestamp	13-06 18:45
Backup age (RPO exposure)	1 hr 15 min
T_break	N/A
T_detect	N/A
T_start	20:01
T_done	20:01

Field	Value
RPO measured	1 hr 15 min
RTO measured	4 sec
Post-flight	<code>SELECT 1</code> returned 1 row
Result	PASS

Service 3 — OwnCloud

Drill 1 — Operator: Ed

Field	Value
T0	13-06 03:46
Latest backup timestamp	13-06 03:42
Backup age (RPO exposure)	4 min
T_break	03:49:20 — <code>mv /var/www/owncloud/data /var/www/owncloud/data.bak</code>
T_detect	03:49:20 — manual observation
T_start	03:49:20
T_done	03:49:21
RPO measured	7 min
RTO measured	1 min
Post-flight	<code>occ status</code> — FAIL (data directory not in backup scope)
Result	FAIL

Drill 2 — Operator: Logan

Field	Value
T0	13-06 10:05
Latest backup timestamp	13-06 06:00
Backup age (RPO exposure)	4 hr 5 min
T_break	10:07
T_detect	10:07
T_start	10:07
T_done	10:08

Field	Value
RPO measured	4 hr 5 min
RTO measured	1 min
Post-flight	occ status — FAIL (data directory not in backup scope)
Result	FAIL

Drill 3 — Operator: Ed

Field	Value
T0	13-06 20:00
Latest backup timestamp	13-06 18:00
Backup age (RPO exposure)	2 hr
T_break	20:02
T_detect	20:02
T_start	20:02
T_done	20:03
RPO measured	2 hr 2 min
RTO measured	1 min
Post-flight	occ status — FAIL (data directory not in backup scope)
Result	FAIL

6. Metrics Summary

Service	Target RPO	Worst Measured RPO	Target RTO	MTTR (avg RTO)	Pass/Fail
Host config	24 hr	4 hr 6 min	10 min	2 min	PASS
MariaDB	24 hr	3 hr 20 min	15 min	4 sec	PASS
OwnCloud	24 hr	4 hr 5 min	20 min	1 min	FAIL

All measured RPO and RTO values are well within targets for Host Config and MariaDB. OwnCloud fails not on timing but on restore completeness.

7. Expected vs Actual

Host Config

Expected RTO ≤ 10 min and RPO ≤ 24 hr. All three drills completed in approximately 2 minutes with no errors. The rsync restore is fast because the `/etc/` tree is small. Reality matched the plan — this service is well covered by the current backup system.

MariaDB

Expected RTO ≤ 15 min. All three drills completed in under 5 seconds. The 2.5MB dump restores almost instantly on backup-d. RPO was well within the 24-hour target across all drills. Reality significantly exceeded the plan — RTO target was conservative. The main future risk is RTO degradation as the database grows.

OwnCloud

Expected RTO ≤ 20 min and full service recovery. All three drills failed post-flight validation. The rsync backup covers `/var/www/owncloud/config/` only — the user data directory (`/var/www/owncloud/data/`) was not included in scope during Lab 11.1. The `occ status` check requires both directories to be present. Config restores successfully in under 1 minute, but a full OwnCloud recovery is not currently possible from backup alone. This is a critical gap that must be addressed before this service can be considered protected.

8. Evidence Appendix

Host Config — Drill 1 rsync output (excerpt)

```
sending incremental file list
sent 6,873 bytes  received 241,082 bytes  165,303.33 bytes/sec
total size is 955,940  speedup is 3.86
T_done: Sat Jun 13 03:48:58 2026
```

MariaDB — Drill 1 restore output

```
T_start: Sat Jun 13 03:47:17 2026
T_done:  Sat Jun 13 03:47:21 2026
+---+
| 1 |
+---+
| 1 |
+---+
PASS
```

OwnCloud — Drill 1 post-flight output

```
Your Data directory is invalid
Please check that the data directory contains a file ".ocdata" in its root.
An unhandled exception has been thrown:
```

```
Exception: Environment not properly prepared.
FAIL
```

9. Risks & Weaknesses

Risk 1 — OwnCloud data directory not in backup scope (Critical)

The most significant finding from this validation. OwnCloud user uploads in `/var/www/owncloud/data/` are not backed up. A server failure would result in permanent loss of all user files. Config can be restored but the service cannot be returned to a working state without the data directory.

Mitigation: Add `/var/www/owncloud/data/` to `backup-app-d.sh`. OwnCloud must be put into maintenance mode before the rsync to avoid partial file copies during active uploads. Follow-up ticket required.

Risk 2 — No automated restore verification

Cron confirms the backup script ran and rsync exited 0, but nothing verifies the dump or snapshot is actually restorable until a human tries. A silent corruption or truncated dump would not be detected until a real recovery attempt.

Mitigation: Add a weekly automated test restore job that imports the latest MariaDB dump into a throwaway database and checks `SELECT COUNT(*) > 0` from a known table.

Risk 3 — backup-d is a single point of failure

All backups from all three hosts land on backup-d only. If backup-d fails (disk failure, accidental deletion, ransomware), all backups are lost simultaneously.

Mitigation: Add an off-host secondary — rsync or rclone to an off-site target or object storage. Out of scope for this lab but a critical production gap.

Risk 4 — MariaDB RTO will degrade with data growth

The current 2.5MB dump restores in 4 seconds. At 1GB+ the same procedure could take 10+ minutes, approaching or exceeding the 15-minute RTO target.

Mitigation: Monitor dump size over time. Consider incremental or binary log-based backup strategy before the dataset grows significantly.

10. Recommendations

Action	Effort	Priority
Add OwnCloud data/ to backup scope with maintenance mode	Low (1–2 hrs)	Critical
Add automated weekly restore verification job	Medium (2–4 hrs)	High
Add off-host secondary backup destination	Medium (2–4 hrs)	High

Action	Effort	Priority
Monitor MariaDB dump size and set RTO alert threshold	Low (1 hr)	Medium